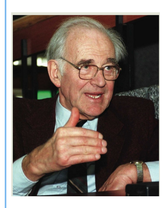
Modeling Isotopic Fractionation with Particle-Based Microphysics using the Bolin Number

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Motivation

The isotopic composition of precipitation and cloud water reflects the history of both diffusional and collisional growth of the hydrometeors. Stable water isotopologues (e.g., HDO) provide valuable insight into the sources and pathways of water in the environment, making them natural tracers of the hydrological cycle. Diffusional and phase-change processes drive isotopic fractionation through both equilibrium and kinetic effects, with kinetic fractionation coupled with ventilation and saturation dynamics. By constraining supersaturation and mixing during entrainment, modeling isotopic fractionation enables reconstruction of saturation conditions and cloud processes and could help distinguish between inhomogeneous and homogeneous mixing.

Bert Bolin (1925-2008)



Swedish climate scientist and the first chairman of the Intergovernmental Panel on Climate Change (IPCC). The timescale ratio, named the Bolin number herein, was defined in Bolin (1958) in the context of tritium content in precipitation.

Brilliant yet humble, Bert Bolin was an excellent communicator, a leader despite his natural shyness, and one who always gave credit to others rather than to himself. – Watson (2008)

Water isotopologues

We consider water $\mathrm{H_2O}$ and single-substituted stable isotopologues: $^1\mathrm{H}^2\mathrm{HO}$, $\mathrm{H_2^{17}O}$ and $\mathrm{H_2^{18}O}$

- $\alpha(T)$ – equilibrium fractionation factor, T – temperature,
- $R_{\mathrm{phase}}^{\mathrm{isotope}} = \frac{n'}{n}$ – heavy to light isotopic ratio: $R_{\mathrm{vap}}, R_{\mathrm{liq}}$,
- $\delta^{\mathrm{isotope}} = \frac{R^{\mathrm{isotope}}}{R_{\mathrm{SMOW}}} - 1$

In the isotopic equilibrium $R_{\mathrm{liq}} = \alpha(T) R_{\mathrm{vap}}$.

Isotopologue diffusion timescales

We consider e-folding timescales for the diffusion processes, separately for light and heavy isotopologues:

$$\tau = \left(\frac{1}{m} \frac{\mathrm{d}m}{\mathrm{d}t} \right)^{-1}, \tag{1}$$

where m is a mass of specific isotopologues in the droplet. We use m' notation for heavy isotopologues, and m for total mass of water in a droplet.

We introduce the **Bolin number** (3), and define e-folding timescale for the heavy isotopologue as

$$\tau' = \mathrm{Bo} \left(\frac{1}{m} \frac{\mathrm{d}m}{\mathrm{d}t} \right)^{-1}. \tag{2}$$

Similar timescales can be found in the literature, but there is no agreement on one form (see Table 1).

Particle based cloud microphysics modeling

We use Lagrangian approach for modeling droplets population tracking attributes such as: water mass, nucleus mass, heat content, etc. These need to be extensive quantities to streamline handling of collisions. We extend it with a new attribute: **isotopic content of the droplet** (n' – number of moles of the isotopologue). The environmental properties (e.g., relative humidity, temperature, water vapor mixing ratio) are in Eulerian approach treated as a field. The **Super Droplet Method** (Shima et al. 2009) allows to numerically track droplets undergoing collisional growth. This method is widely used in atmospheric modeling (see Morrison et al. (2026)).

Table 1: τ' in literature

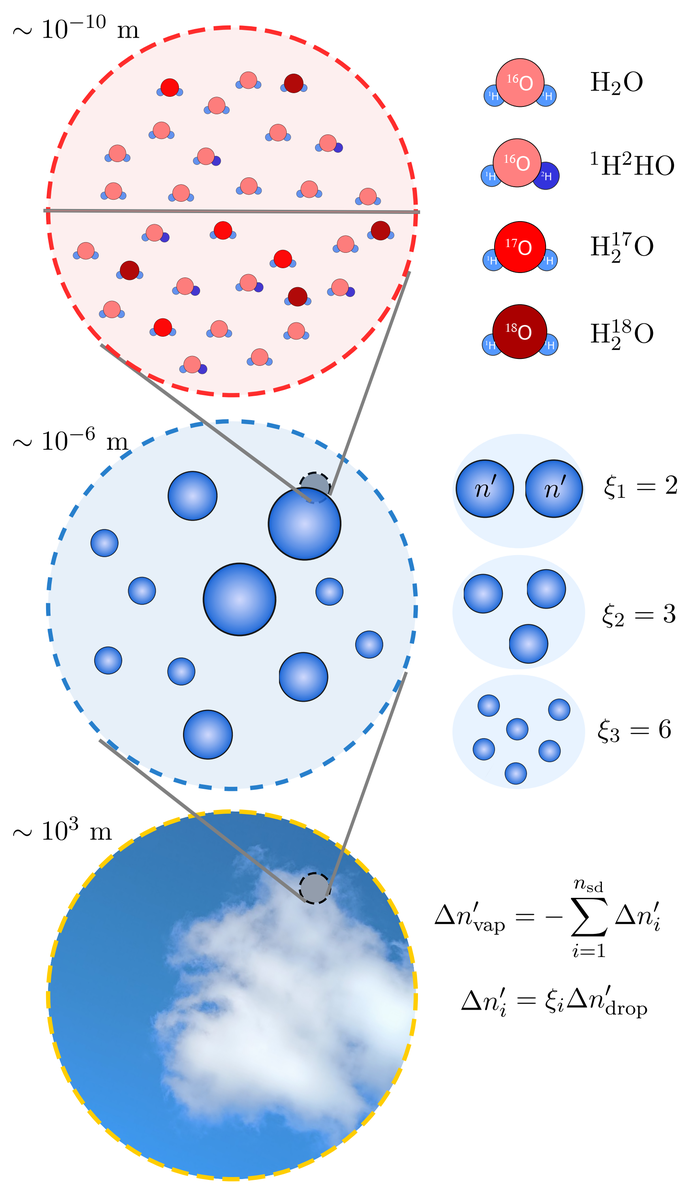
Reference	Relaxation timescale
Bolin (1958) (eq. 5) $c_1 = \alpha\rho_{\mathrm{s}}/\rho$; for tritium only	$\left(c_1 \frac{1}{m} \frac{\mathrm{d}m}{\mathrm{d}t} \right)^{-1}$
Friedman, Machta, and Soller (1962) (eq. 2) A – experimentally determined ventilation factor	$\frac{4\pi r^2}{3\alpha\rho_s(T_\circ) D A}$
Booker (1965) (eq. 1) N = one-quarter of the absolute mean speed of water-vapour molecules; for tritium	$\frac{r^2\rho_{\mathrm{liq}}}{3\alpha f D \rho_{\mathrm{vap}}} \left(1 + \frac{D}{\alpha r N} \right)$
Miyake, Matsubaya, and Nishihara (1968) (eq. 28)	$\frac{r^2 \alpha \rho_{\mathrm{s}} R^*_{\mathrm{vap}} T}{3 e_{\mathrm{s}} f D}$
Stewart (1975) (eq. 7) n – an experimental turbulence parameter	$\frac{\alpha m R T}{4\pi r e_{\mathrm{s}} k (D')^n}$
Jouzel (1986) (eq. 7)	$\frac{\alpha r^2 \rho_{\mathrm{liq}}}{3 D' f' \rho_{\mathrm{s}}}$
Gedzelman and Arnold (1994) (eq. 24) For $S = 1$	$\frac{\alpha r^2 \rho_{\mathrm{liq}}}{3 D_{\mathrm{i}} f_{\mathrm{i}} \rho_{\mathrm{s}}}$
Bolot, Legras, and Moyer (2013) (eq. B5) $A = (1 + f D F_{\mathrm{k}})^{-1}$	$\frac{\frac{r^2 \rho_{\mathrm{liq}} \alpha}{3\rho_{\mathrm{s}} D' f'}}{(S + A(1-S)) - \alpha \frac{Df}{D'f'} A (1-S)}$
Hiron and Flossmann (2020) (eq. 13)	$\frac{\alpha_k r^2 \rho_{\mathrm{liq}}}{3 D_i f_i \rho_s}$

- $D_{\mathrm{rat}} = \frac{D'(T)}{D(T)}$ – diffusivity ratios
- α – isotope equilibrium fractionation factor
- $F_{\mathrm{k}}, F_{\mathrm{d}}$ – heat-conduction/vapor-diffusion terms (see, Rogers and Yau (1996))
- m, m' – heavy/light isotopologue mass
- RH – relative humidity

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Bolin number

$$\mathrm{Bo} := \frac{\tau'}{\tau} = \left(\frac{1}{m} \frac{\mathrm{d}m}{\mathrm{d}t} \right) \cdot \left(\frac{1}{m'} \frac{\mathrm{d}m'}{\mathrm{d}t} \right)^{-1} \tag{3}$$

$$\mathrm{Bo} = \frac{\alpha}{D_{\mathrm{rat}}} \underbrace{(1 - \mathrm{RH})}_{=0} \left(\mathrm{RH} \left(1 + \frac{F_{\mathrm{k}}}{F_{\mathrm{d}}} \right) \underbrace{\left(1 - \alpha \frac{R_{\mathrm{vap}}}{R_{\mathrm{liq}}} \right)}_{=0} + 1 - \mathrm{RH} \right)^{-1}$$

phase equilibrium; isotopic equilibrium

Isotopic content dynamics and numerics

Let n', n be the in-droplet number of moles of heavy and light isotopologues, and $n'_{\mathrm{vap}}/m_{\mathrm{dry}}$ the molality in ambient air.

To facilitate numerical solution, leveraging the **positive definiteness** of the number of moles in the droplets, we integrate the diffusion process using $x = \log(n')$:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{n'} \frac{\mathrm{d}n'}{\mathrm{d}t}. \tag{4}$$

We have

$$x_{\mathrm{new}} = x_{\mathrm{old}} + \Delta t \frac{\mathrm{d}x}{\mathrm{d}t},$$

$$\log(n'_{\mathrm{new}}) = \log(n'_{\mathrm{old}}) + \Delta t \frac{1}{n'} \frac{\mathrm{d}n'}{\mathrm{d}t}.$$

For each super-droplet calculate new number of moles of isotopologues from the mass total change (given by condensation):

$$\frac{\mathrm{d}n'}{\mathrm{d}t} = \mathrm{Bo}^{-1} \frac{n'}{n} \frac{\mathrm{d}n}{\mathrm{d}t} \tag{5}$$

We use **Euler method implicit in attribute and explicit in the ambient field**, taking ambient air isotopic content (R_{vap}) from the previous step. The Bolin number (Bo) is recalculated with respect to new R_{liq}.

Let $f(\beta) = \mathrm{Bo}^{-1}(\beta)$, where $\beta = R_{\mathrm{vap}}/R_{\mathrm{liq}}$:

$$f(\beta) = \frac{\left(1 + \frac{F_k}{F_d}\right) \mathrm{RH} \, (1 - \alpha\beta) - \mathrm{RH} + 1}{\alpha / D_{\mathrm{rat}} (1 - \mathrm{RH})} \tag{6}$$

$$f'(\beta) = \frac{D_{\mathrm{rat}} \left(1 + \frac{F_k}{F_d}\right) \mathrm{RH}}{\mathrm{RH} - 1} =: A \tag{7}$$

$$f(\beta_{\mathrm{new}}) = f(\beta_{\mathrm{old}}) + (\beta_{\mathrm{new}} - \beta_{\mathrm{old}}) A \tag{8}$$

Using (8) to (5):

$$\frac{\mathrm{d}n'}{dt} = f(\beta_{\mathrm{new}}) \frac{n'_{\mathrm{new}}}{n_{\mathrm{new}}} \frac{dn}{dt} \tag{9}$$

PySDM `open-atmos.github.io/PySDM/`

The concept is implemented in the PySDM package, an open-source Pythonic implementation of the Super-Droplet Method for simulating the dynamics of population of particles. Isotopic fractionation is introduced as a new "dynamic" within the package. Enabling it does not change the ordinary diffusional growth logic (computed using mean isotopic composition). Moreover, it is fully compatible with the existing collision and breakup dynamics.

Revisit of the Gedzelman and Arnold (1994) analysis

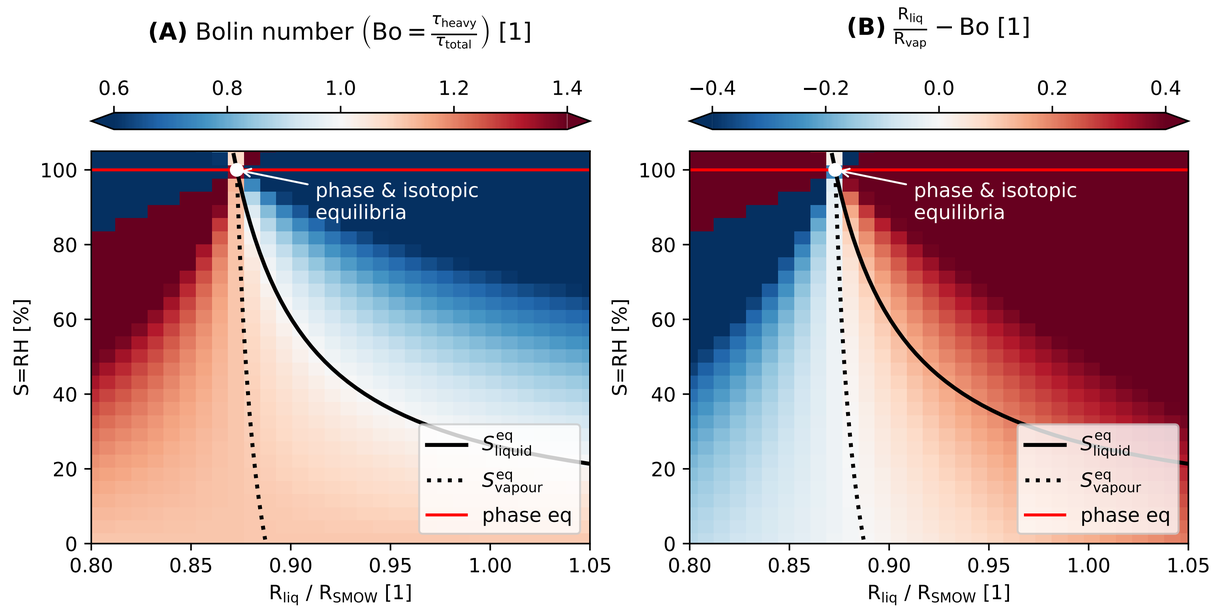


Figure: The Bolin number (A) equals 1 when the isotopic composition of liquid droplets is not changing $\left(\text{i.e., } \frac{\mathrm{d}R_{\mathrm{liq}}}{\mathrm{d}t} = 0\right)$. For the isotopic composition of ambient vapor (B) to be in equilibrium $\left(\text{i.e., } \frac{\mathrm{d}R_{\mathrm{vap}}}{\mathrm{d}t} = 0\right)$, the following condition must be met: $\left(\frac{R_{\mathrm{liq}}}{R_{\mathrm{vap}}} - \mathrm{Bo} = 0\right)$. The isotopologue considered in both plots is $^1\mathrm{H}^2\mathrm{H}^{16}\mathrm{O}$. Black lines: isotopic equilibrium curves from Gedzelman and Arnold 1994 (Fig. 2); red line: phase equilibrium ($\mathrm{RH} = 100\%$); white dot: phase and isotopic equilibrium, $R_{\mathrm{liq}} = \alpha(T) R_{\mathrm{vap}}$. The environmental setting follows that of the paper: $T = 10°\mathrm{C}$, $\delta_{^2\mathrm{H}} = \frac{R_{\mathrm{vap}}}{R_{\mathrm{SMOW}}} - 1 = -200‰$.

Figure is created inside the Jupyter Notebook (`.ipynb`) and available on GitHub.

`.../Gedzelman_and_Arnold_1994/fig_2.ipynb`

The isotopic extension to the PySDM package is carried as one of the projects in the group `open-atmos-krk`.

`open-atmos-krk.github.io/projects/sdm_isotopes`

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